

LIVINGSTON COUNTY AP, MI, USA

WMO: 725378

Lat: 42.629N

Lon: 83.984W

Elev: 288

StdP: 97.92

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04887

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.5	(c) -14.9	(d) -22.1	(e) 0.5	(f) -16.0	(g) -19.1	(h) 0.7	(i) -13.4	(j) 10.9	(k) -3.1	(l) 9.4	(m) -1.7	(n) 2.4	(o) 230	(p) 0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.3	22.2	30.9	21.7	28.9	20.8	24.2	29.1	23.3	27.9	22.4	26.8	4.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.1	26.5	22.1	17.4	26.0	21.1	16.3	24.9	74.5	29.3	70.7	27.8	67.3	26.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.1	7.2	-21.0	34.3	3.5	1.5	-23.5	35.4	-25.5	36.3	-27.5	37.1	-30.1	38.3
			-21.1	26.1	3.5	1.8	-23.6	27.4	-25.6	28.4	-27.6	29.4	-30.1	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-4.1	-3.1	1.8	8.5	14.8	20.0	22.1	21.1	17.5	10.9	4.6	-1.2
	DBStd	10.52	6.21	6.06	6.20	5.14	4.75	3.82	3.18	3.08	4.29	4.86	5.11	5.34
	HDD10.0	1739	437	367	264	87	11	0	0	0	2	48	176	348
	HDD18.3	3594	694	599	512	296	132	26	6	10	66	236	413	605
	CDD10.0	1551	1	1	12	42	158	301	376	344	227	76	13	2
	CDD18.3	364	0	0	1	2	22	76	123	96	41	5	0	0
	CDH23.3	3257	0	0	6	28	220	697	1141	764	358	43	1	0
	CDH26.7	1002	0	0	1	4	52	232	392	210	104	7	0	0
(14) Wind	WSAvg	3.1	3.7	3.6	3.5	3.7	3.1	2.6	2.3	2.2	2.3	3.0	3.4	3.5
(15) Precipitation	PrecAvg	845	55	47	54	68	89	95	83	78	78	71	62	56
	PrecMax	1044	102	92	98	145	191	165	165	154	235	186	113	98
	PrecMin	610	17	20	19	11	28	32	39	17	15	7	27	24
	PrecStd	111	27	23	22	26	42	36	38	32	51	39	25	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.9	23.0	27.0	31.0	32.9	34.1	32.8	32.3	27.6	20.2	15.2
		MCWB	11.1	10.8	15.0	16.4	20.1	22.6	23.2	22.3	21.4	20.1	13.7	13.6
	2%	DB	9.9	11.2	18.5	22.9	27.9	31.6	32.4	30.9	29.7	24.0	17.2	12.0
		MCWB	8.2	8.3	13.4	13.8	18.8	21.7	22.7	21.9	20.8	17.1	12.4	9.9
	5%	DB	7.1	7.6	14.7	20.3	26.0	29.4	30.9	28.9	27.4	22.0	15.0	8.3
		MCWB	5.3	4.8	10.5	12.7	18.0	20.6	22.1	21.2	19.6	16.3	11.3	6.5
	10%	DB	3.8	4.7	11.4	17.7	23.3	27.4	28.8	27.5	25.1	19.0	12.5	5.9
		MCWB	2.4	2.5	7.7	11.1	16.7	19.8	21.0	20.3	18.6	14.7	9.3	3.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.4	16.5	18.7	22.2	24.5	26.0	24.9	23.8	21.4	15.9	13.8
		MCDB	12.9	14.0	21.9	24.4	27.3	30.7	30.6	30.4	29.1	26.2	18.6	15.4
	2%	WB	8.4	8.4	13.9	16.0	20.6	23.1	24.7	23.9	22.4	19.2	13.9	10.0
		MCDB	9.3	10.3	17.4	20.5	25.5	28.6	29.4	28.2	26.6	22.3	16.2	11.3
	5%	WB	5.1	5.0	11.2	14.0	19.4	22.2	23.5	22.8	21.2	17.1	11.8	6.7
		MCDB	6.8	7.3	14.3	19.0	24.0	27.4	28.1	26.6	24.8	20.2	14.4	8.1
	10%	WB	2.6	2.7	7.8	12.1	17.8	21.1	22.5	21.8	19.9	15.2	9.5	4.0
		MCDB	3.9	4.7	10.6	16.9	22.0	25.6	27.0	25.4	23.9	18.2	12.4	5.6
(35) Mean Daily Temperature Range	MDBR		7.2	8.1	9.5	11.2	11.4	11.2	11.6	11.0	11.5	10.1	8.1	6.5
	5% DB	MCDBR	9.1	10.7	13.8	14.9	13.5	13.2	13.0	12.4	13.6	13.4	11.3	9.2
		MCWBR	8.0	8.3	9.1	8.4	6.7	5.7	5.5	5.2	6.0	7.5	7.9	7.6
	5% WB	MCDBR	8.9	10.1	12.7	13.0	11.7	11.4	10.9	10.1	11.2	11.4	10.3	8.6
		MCWBR	8.2	8.4	9.1	8.9	6.9	6.0	5.8	5.3	6.0	7.6	8.3	7.8
(40) Clear-Sky Solar Irradiance	taub		0.295	0.333	0.343	0.386	0.413	0.435	0.425	0.418	0.380	0.346	0.315	0.298
	taud		2.372	2.241	2.357	2.258	2.236	2.205	2.248	2.259	2.353	2.442	2.481	2.412
	Ebn at Noon		843	856	899	880	863	841	846	839	848	839	819	807
	Edh at Noon		80	108	109	130	137	142	134	129	109	88	72	71
(44) All-Sky Solar Radiation	RadAvg		1.50	2.37	3.56	4.58	5.34	6.03	6.10	5.24	4.22	2.72	1.69	1.18
	RadStd		0.13	0.30	0.37	0.41	0.48	0.38	0.29	0.22	0.39	0.25	0.22	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.54	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page